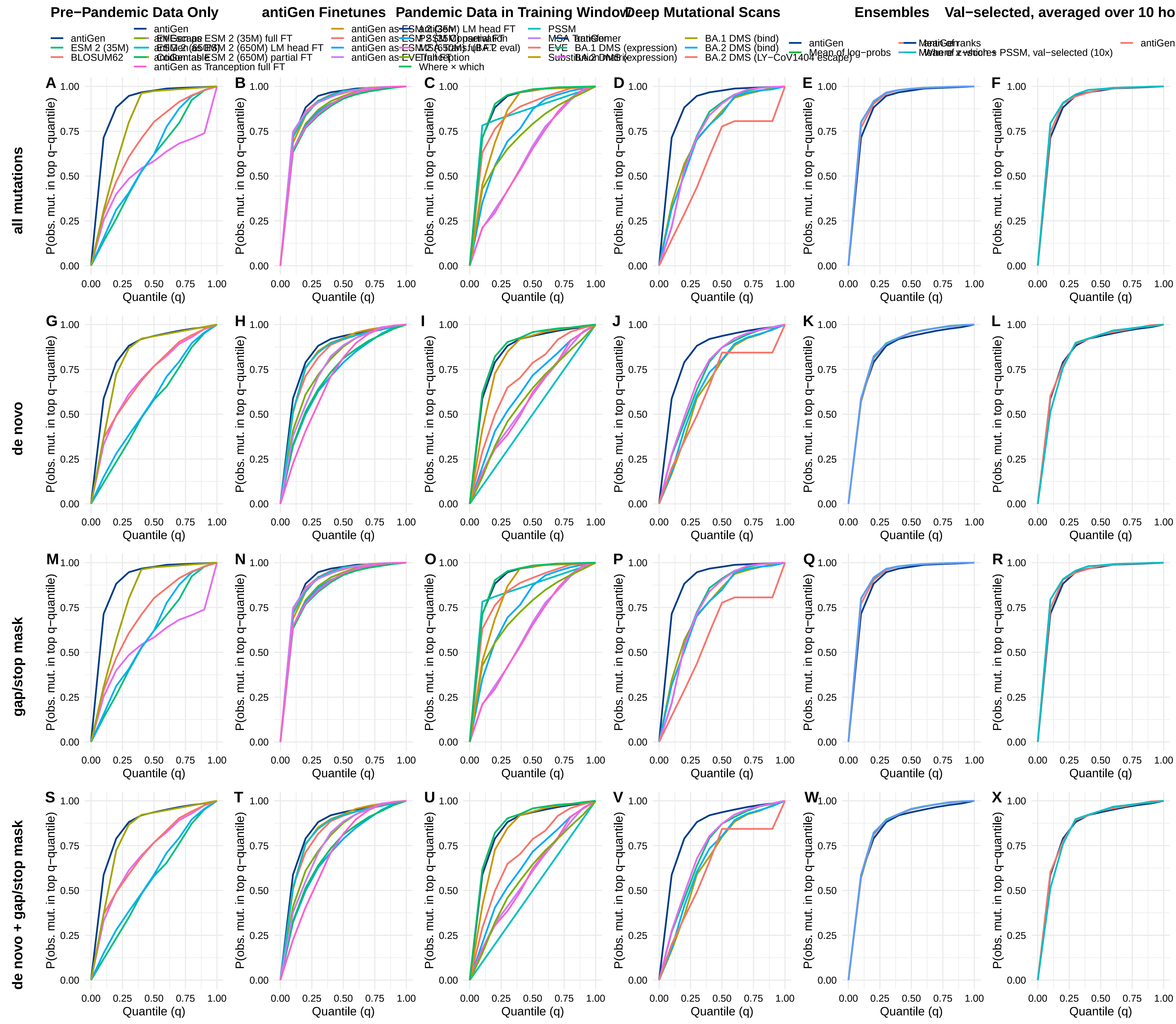
Pre−Pandemic Data Only
antiGen Finetunes
Pandemic Data in Training Window
Deep Mutational Scans
Ensembles
Val−selected, averaged over 10 ho
antiGen
ESM 2 (35M)
BLOSUM62
antiGen
EVEscape
ESM 2 (650M)
Codon table
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) LM head FT
antiGen as ESM 2 (650M) partial FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (650M) full FT
antiGen as EVE full FT
antiGen
Position conservation
MSA Transformer (BA.2 eval)
Tranception
Where × which
PSSM
MSA Transformer
EVE
Substitution matrix
antiGen
BA.1 DMS (expression)
BA.2 DMS (expression)
BA.1 DMS (bind)
BA.2 DMS (bind)
BA.2 DMS (LY−CoV1404 escape)
antiGen
Mean of log−probs
Mean of ranks
Mean of z-scores
antiGen
Where × which + PSSM, val−selected (10x)
antiGen
all mutations
de novo
gap/stop mask
de novo + gap/stop mask
A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
P(obs. mut. in top q−quantile)
1.00
0.75
0.50
0.25
0.00
Quantile (q)
0.00
0.25
0.50
0.75
1.00